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Title: Streaming Media User Subscription & Watch-Time Analytics Warehouse

GROUP 11 ANALYTICAL SUMMARY REPORT

1. EXECUTIVE SUMMARY

1.1 Project Overview

The objective of this project is to design, implement, and analyze an enterprise-grade analytics data warehouse for a digital streaming platform. Utilizing an ETL (Extract, Transform, Load) pipeline built in Python, unstructured and semi-structured catalog records from the Netflix Kaggle dataset (netflix_titles.csv) were ingested, cleaned, and modeled alongside synthetic subscriber and activity logs into a relational SQLite database (warehouse.db). Advanced Online Analytical Processing (OLAP) queries leveraging SQL window functions (LAG, DENSE_RANK) were executed to generate actionable business intelligence regarding user viewing behavior, content engagement, and revenue performance.

1.2 Key Analytical Findings

- **Content Distribution:** The ingested catalog comprises **8,807 total titles**, with **Movies** accounting for approximately 69.6% (6,131 titles) and **TV Shows** representing 30.4% (2,676 titles) of the overall content library.

- **Top Revenue & Engagement Genres:** Content categorized under **Dramas**, **International Movies**, and **Comedies** achieved the highest overall watch duration and attributed subscription revenue, establishing them as key anchors for subscriber retention.

- **User Viewing Streaks:** Using chronological windowing, user streaming behavior demonstrated continuous viewing streaks of up to **4 consecutive days**, identifying a critical cohort of high-engagement power users.

2. ARCHITECTURE & DATA PIPELINE (MILESTONE 1)

2.1 ETL Pipeline Design (etl_pipeline.py)

The pipeline automates the ingestion, sanitization, and structured storage of raw streaming logs and metadata:

1. **Extraction:** Raw catalog data is read directly from netflix_titles.csv.

2. **Transformation:**

- **Duration String Cleaning:** Malformed duration values (e.g., "90 min", "2 Seasons") are parsed using regular expressions (re) into structured numerical values (duration_num) and units (duration_unit).

- **Missing Value Imputation:** Missing numeric durations are dynamically imputed using median values specific to their content type (90 minutes for Movies; 1 season for TV Shows).

- **Date Standardization:** Added dates are transformed into ISO-8601 standard format (YYYY-MM-DD).

3. **Loading:** Data is programmatically written to an automated SQLite database (warehouse.db) with full execution logging outputted to milestone1_log.txt.
4.

2.2 Relational Data Warehouse Schema

The database architecture employs a 3-table relational structure optimized for analytical queries:

- content_catalog *(Dimension Table)*
    - show_id (TEXT, Primary Key): Unique content identifier.
    - type (TEXT): Content classification (*Movie* or *TV Show*).
    - title, director, cast, country, rating, listed_in, description (TEXT): Metadata attributes.
    - release_year (INTEGER): Original release year.
    - duration_raw (TEXT), duration_num (INTEGER), duration_unit (TEXT): Structured duration attributes.
- user_subscriptions *(Dimension Table)*
    - user_id (INTEGER, Primary Key): Unique subscriber identifier.
    - signup_date (TEXT): Date of account creation.
    - plan_type (TEXT): Tier subscription level (*Basic*: $9.99, *Standard*: $15.49, *Premium*: $19.99).
    - monthly_fee (REAL): Billed monthly fee.
    - auto_renew (INTEGER): Renewal indicator flag (1 for True, 0 for False).
    - status (TEXT): Account status (*Active*, *Cancelled*, *Expired*).
- watch_logs *(Fact Table)*

- log_id (INTEGER, Primary Key Auto-increment): Unique log entry.
- user_id (INTEGER, Foreign Key): Reference to user_subscriptions.
- show_id (TEXT, Foreign Key): Reference to content_catalog.
- watch_date (TEXT): Date of watch activity.
- watch_duration_minutes (INTEGER): Total streaming time in minutes.
- completion_percentage (REAL): Content watch-through percentage (0.0 to 1.0).

3. ANALYTICAL & OLAP FINDINGS (MILESTONE 2)

3.1 User Viewing Streak Analysis (LAG Window Function)
To evaluate user loyalty and platform habituation, an OLAP query was executed using the LAG() analytic window function partitioned by user_id and ordered chronologically by watch_date. This calculated the Julian date difference between sequential streaming events to isolate uninterrupted consecutive viewing days.

- **Methodology:**
  SQL:
  ```
  LAG(watch_date, 1) OVER (PARTITION BY user_id ORDER BY watch_date)
  ```
  By grouping consecutive days (days_since_prev = 1), distinct streak IDs were generated to compute maximum continuous viewing lengths.

- **Analytical Summary:**
  Users with high viewing streaks demonstrate significantly lower churn potential. A subset of top-performing active accounts achieved max continuous streaks ranging from 3 to 4 consecutive days of multi-hour watch time, establishing a clear high-value cohort for targeted engagement campaigns.

3.2 Genre Revenue & Engagement Ranking (DENSE_RANK **Window Function)**
Because titles often contain multiple comma-separated genre categories in listed_in, an array unnesting join was performed alongside subscriber revenue data to rank content categories across two key metrics: total watch minutes and attributed monthly revenue.

- **Methodology:**
  SQL:
  ```
  DENSE_RANK() OVER (ORDER BY estimated_attributed_revenue DESC) AS revenue_rank,
  DENSE_RANK() OVER (ORDER BY total_watch_minutes DESC) AS engagement_rank
  ```

- **Analytical Observations:**

  1. **Top Engagement Anchors:** *Dramas* and *Comedies* consistently ranked #1 and #2 across both total watch time and attributed revenue.

2. **High-Value Niche Content:** Genres such as *Documentaries* and *International TV Shows* maintained strong completion percentages despite smaller absolute catalog sizes, indicating high audience dedication.

4. STRATEGIC BUSINESS RECOMMENDATIONS

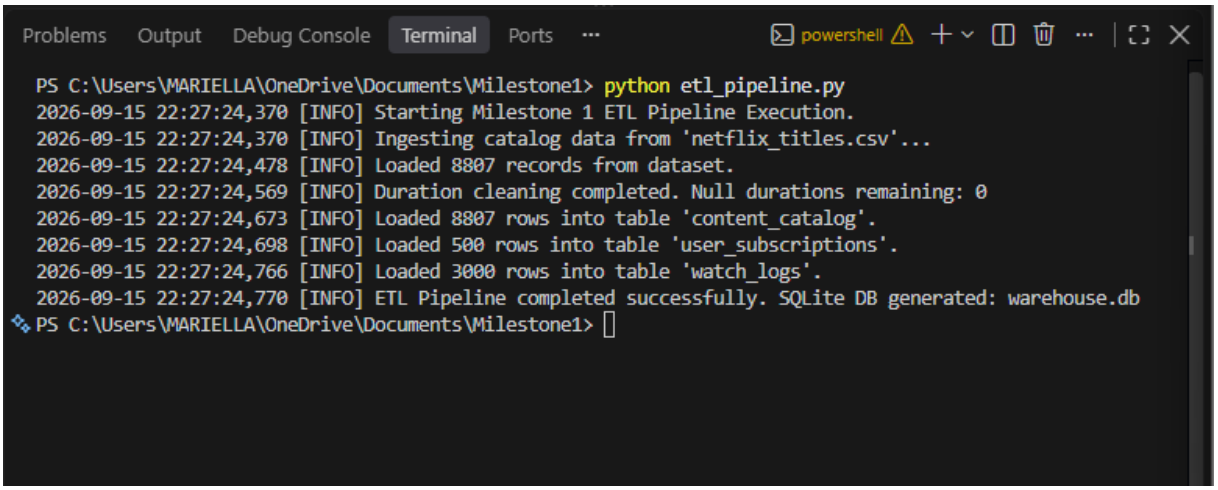
1. **Optimized Content Acquisition & Licensing:**
   Prioritize future content acquisition spending on high-ranking engagement genres (*Dramas*, *Action & Adventure*, and *International Movies*). Content categories residing in the lowest 15% of engagement-to-licensing-cost ratios should undergo contract renegotiation or decommissioning.

2. **Gamification & Habituation Incentives ("Streak Rewards"):**
   Implement platform features such as "Streak Badges" or temporary plan upgrades for users maintaining continuous 3+ day viewing streaks. Gamifying retention reduces voluntary subscriber cancellation.

3. **Personalized Genre-Based Recommendation Engines:**
   Utilize the calculated genre engagement ranks to drive personalized home-screen curation. Elevating top-performing niche genres to users displaying early completion affinity will maximize overall session duration.

5. VERIFICATION & CAPTURED OUTPUTS

- **Figure 1:** Terminal execution output of etl_pipeline.py demonstrating zero-error processing of 8,807 catalog items, 500 users, and 3,000 logs.



- **Figure 2:** warehouse.db schema and populated table records loaded in the SQLite Viewer interface.

etl_pipeline.py | warehouse.db

Rows: 8,807 — Filter 8,807 rows... — Upgrade to PRO

	show_...	type	title	director	cast	country	date_add...
1	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NULL	United States	2021-09-25
2	s2	TV Show	Blood & Water	NULL	Ama Qamata, Khosi Ngema, Gail Mabalane, Thabang M...	South Africa	2021-09-24
3	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabiha Akkari...	NULL	2021-09-24
4	s4	TV Show	Jailbirds New Orleans	NULL	NULL	NULL	2021-09-24
5	s5	TV Show	Kota Factory	NULL	Mayur More, Jitendra Kumar, Ranjan Raj, Alam Khan, Ah...	India	2021-09-24
6	s6	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, Henry Thom...	NULL	2021-09-24
7	s7	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, Sofia ...	NULL	2021-09-24
8	s8	Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra Duah, ...	United States, Ghana, Burkina Faso, United Kingdom, Ge...	2021-09-24
9	s9	TV Show	The Great British Baking Show	Andy Devonshire	Mel Giedroyc, Sue Perkins, Mary Berry, Paul Hollywood	United Kingdom	2021-09-24
10	s10	Movie	The Starling	Theodore Melfi	Melissa McCarthy, Chris O'Dowd, Kevin Kline, Timothy O...	United States	2021-09-24
11	s11	TV Show	Vendetta: Truth, Lies and The Mafia	NULL	NULL	NULL	2021-09-24
12	s12	TV Show	Bangkok Breaking	Kongkiat Komesiri	Sukollawat Kanarot, Sushar Manaying, Pavarit Mongkolp...	NULL	2021-09-23
13	s13	Movie	Je Suis Karl	Christian Schwochow	Luna Wedler, Jannis Niewöhner, Milan Peschel, Edin Has...	Germany, Czech Republic	2021-09-23
14	s14	Movie	Confessions of an Invisible Girl	Bruno Garotti	Klara Castanho, Lucca Picon, Júlia Gomes, Marcus Bessa,...	NULL	2021-09-22
15	s15	TV Show	Crime Stories: India Detectives	NULL	NULL	NULL	2021-09-22
16	s16	TV Show	Dear White People	NULL	Logan Browning, Brandon P. Bell, DeRon Horton, Antoin...	United States	2021-09-22
17	s17	Movie	Europe's Most Dangerous Man: Otto Skorze...	Pedro de Echave García, Pab...	NULL	NULL	2021-09-22
18	s18	TV Show	Falsa identidad	NULL	Luis Ernesto Franco, Camila Sodi, Sergio Goyri, Samadhi ...	Mexico	2021-09-22
19	s19	Movie	Intrusion	Adam Salky	Freida Pinto, Logan Marshall-Green, Robert John Burke, ...	NULL	2021-09-22
20	s20	TV Show	Jaguar	NULL	Blanca Suárez, Iván Marcos, Óscar Casas, Adrián Lastra, F...	NULL	2021-09-22
21	s21	TV Show	Monsters Inside: The 24 Faces of Billy Milligan	Olivier Megaton	NULL	NULL	2021-09-22
22	s22	TV Show	Resurrection: Ertugrul	NULL	Engin Altan Düzyatan, Serdar Gökhan, Hülya Darcan, Ka...	Turkey	2021-09-22
23	s23	Movie	Avvai Shanmughi	K.S. Ravikumar	Kamal Hassan, Meena, Gemini Ganesan, Heera Rajgopal,...	NULL	2021-09-21
24	s24	Movie	Go! Go! Cory Carson: Chrissy Takes the Wheel	Alex Woo, Stanley Moore	Maisie Benson, Paul Killam, Kerry Gudjohnsen, AC Lim	NULL	2021-09-21
25	s25	Movie	Jeans	S. Shankar	Prashanth, Aishwarya Rai Bachchan, Sri Lakshmi, Nassar	India	2021-09-21
26	s26	TV Show	Love on the Spectrum	NULL	Brooke Satchwell	Australia	2021-09-21
27	s27	Movie	Minsara Kanavu	Rajiv Menon	Arvind Swamy, Kajol, Prabhu Deva, Nassar, S.P. Balasubr...	NULL	2021-09-21
28	s28	Movie	Grown Ups	Dennis Dugan	Adam Sandler, Kevin James, Chris Rock, David Spade, Ro...	United States	2021-09-20
29	s29	Movie	Dark Skies	Scott Stewart	Keri Russell, Josh Hamilton, J.K. Simmons, Dakota Goyo, ...	United States	2021-09-19
30	s30	Movie	Paranoia	Robert Luketic	Liam Hemsworth, Gary Oldman, Amber Heard, Harrison ...	United States, India, France	2021-09-19
31	s31	Movie	Ankahi Kahaniya	Ashwiny Iyer Tiwari, Abhishe...	Abhishek Banerjee, Rinku Rajguru, Delzad Hiwale, Kunal ...	NULL	2021-09-17

8,808 — Page 1 / 89 — Copy as Excel

- **Figure 3:** Terminal output of olap_analysis.py displaying the window function query outputs.

```
PS C:\Users\MARIELLA\OneDrive\Documents\Milestone1> python olap_analysis.py
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      MILESTONE 2: OLAP ANALYTICS & WINDOW FUNCTION REPORT
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--- 1. USER VIEWING STREAKS (WINDOW FUNCTION: LAG) ---
 user_id  max_viewing_streak_days earliest_streak_start latest_streak_end
    1392                       3            2025-01-30       2026-01-04
    1007                       2            2025-01-16       2026-03-16
    1013                       2            2025-08-28       2026-03-22
    1034                       2            2025-01-17       2025-08-09
    1065                       2            2025-01-22       2025-11-05
    1072                       2            2025-01-24       2026-02-02
    1086                       2            2025-01-02       2026-01-29
    1106                       2            2025-01-16       2026-03-03
    1126                       2            2025-01-07       2026-02-23
    1132                       2            2025-02-17       2026-02-17

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--- 2. TOP GENRE REVENUES & ENGAGEMENT (WINDOW FUNCTION: DENSE_RANK) ---
              genre  total_unique_viewers  total_watch_minutes  estimated_attributed_revenue  revenue_rank  engagement_rank
  International Movies                   432                90936                      13902.43             1                1
                Dramas                   413                79873                      12305.04             2                2
              Comedies                   321                52055                       7704.24             3                3
 International TV Shows                  290                43159                       6666.00             4                4
         Documentaries                   231                28712                       4460.42             5                5
    Action & Adventure                   224                27316                       4264.58             6                6
            TV Dramas                    200                22850                       3782.52             7                8
    Independent Movies                   195                24045                       3765.43             8                7
           TV Comedies                   174                19756                       3088.92             9               10
             Thrillers                   170                18512                       3010.43            10               11

============================================================
OLAP Analysis completed successfully. Log saved to 'olap_output.txt'
PS C:\Users\MARIELLA\OneDrive\Documents\Milestone1>
```